Question:

A 35-year-old African American man presents with fever, abdominal pain, and severe weakness since yesterday. On physical examination, the patient is jaundiced and shows a generalized pallor. Past medical history is significant for recently receiving anti-malaria prophylaxis before visiting Nigeria. Laboratory tests show normal glucose-6-phosphate dehydrogenase (G6PD) levels. Peripheral smear shows the presence of bite cells and Heinz bodies. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Autoimmune hemolytic anemia

B. Microangiopathic hemolytic anemia

C. Paroxysmal nocturnal hemoglobinuria (PNH)

D. Glucose-6-phosphate-dehydrogenase (G6PD) deficiency

Reasoning + Answer:

<think> The patient presents with fever, abdominal pain, and severe weakness, along with jaundice and generalized pallor. The patient has a history of recently receiving anti-malaria prophylaxis before visiting Nigeria. The laboratory tests show normal G6PD levels, which is an important clue to rule out G6PD deficiency. The peripheral smear shows the presence of bite cells and Heinz bodies, which are characteristic findings of oxidative hemolysis. The patient's symptoms and laboratory findings are consistent with an acute hemolytic anemia. The presence of Heinz bodies and bite cells suggests that the anemia is due to oxidative damage to the red blood cells. G6PD deficiency is a common cause of acute hemolytic anemia, especially in individuals of African descent, and it is triggered by oxidative stress, which can be caused by infections, medications, or certain foods. The patient's recent travel history and receipt of anti-malaria prophylaxis suggest that the anemia may have been triggered by the medication. The other options are less likely: autoimmune hemolytic anemia typically presents with a positive Coombs test, microangiopathic hemolytic anemia is characterized by schistocytes on the peripheral smear, and paroxysmal nocturnal hemoglobinuria (PNH) is a rare disorder that typically presents with hemolytic anemia, bone marrow failure, and a positive flow cytometry test. </think> <answer> D. Glucose-6-phosphate-dehydrogenase (G6PD) deficiency </answer>